1 of 2 pages

CHAIN OF CUSTODY FORM

(Original Stays With Samples)

SHIP TO: US EPA
ATTN: Sam Smith
26 West Martin Luther King Dr
Cincinnati, OH 45268
Phone: 513-569-7107
smith.samantha@epa.gov

Originator:			
Ship Date:	April 5th 2018	Contact:	
Destination:	Cincinnati	Phone #:	
Recipient:	Sam Smith	E-Mail:	
Phone #:	513-569-7681		

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Sample Description	Container Type	# Containers	Preservative	Analyses Required	Special Instructions for Lab
20180403	12:42:00				1	DI	Field BLK 1					TNH, TRP, TNO2,3, TNOC, TN, TP	
20180403	12:44:00				1	DI	Field BLK 2					"	
20180403	12:45:00				1	DI	BTL BLK 1					"	
20180403	12:46:00				1	DI	BTL BLK 2					"	
20180404	08:38:00	FL	42	0.1	1	SW	UKN					"	
20180404	08:38:00	FL	42	0.1	1	SW	DUP					"	
20180404	09:00:00	FL	44	0.1	1	SW	UKN					"	
20180404	09:36:00	FL	43	0.1	1	SW	UKN					"	
20180404	10:00:00	FL	41	0.1	1	SW	UKN					"	
20180404	10:45:00	FL	04	0.1	1	SW	UKN					"	
20180404	10:57:00	FL	03	0.1	1	SW	UKN					"	
20180404	11:13:00	FL	14	0.1	1	SW	UKN					"	
20180404	11:28:00	FL	15	0.1	1	SW	UKN					"	
20180404	11:53:00	FL	07	0.1	1	SW	UKN					"	
20180404	12:30:00	FL	01	0.1	1	SW	UKN					"	
20180404	13:08:00	FL	05	0.1	1	SW	UKN					"	
20180404	13:18:00	FL	02	0.1	1	SW	UKN					"	
20180405	14:15:00				1	DI	HOUSE DI					"	
20180403	12:42:00				1	DI	Field BLK 1				2% HCl	TOC	
20180403	12:44:00				1	DI	Field BLK 2				"	TOC	
20180403	12:45:00				1	DI	BTL BLK 1				"	TOC	
20180403	12:46:00				1	DI	BTL BLK 2				"	TOC	
20180404	08:38:00	FL	42	0.1	1	SW	UKN				"	TOC	
20180404	08:38:00	FL	42	0.1	1	SW	DUP				"	TOC	
20180404	09:00:00	FL	44	0.1	1	SW	UKN				"	TOC	
20180404	09:36:00	FL	43	0.1	1	SW	UKN				"	TOC	
20180404	10:00:00	FL	41	0.1	1	SW	UKN				"	TOC	

* Matrix Codes: SW=Surface Water, BI=Biotic Invert

Relinquished By	Date:	Time:	Received By	Date:	Time:	Remarks

CHAIN OF CUSTODY FORM
(Original Stays With Samples)

SHIP TO: US EPA
ATTN: Sam Smith
26 West Martin Luther King Dr
Cincinnati, OH 45268
Phone: 513-569-7107
smith.samantha@epa.gov

Originator:			
Ship Date:	April 5th 2018	Contact:	
Destination:	Cincinnati	Phone #:	
Recipient:	Sam Smith	E-Mail:	
Phone #:	513-569-7681		

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Sample Description	Container Type	# Containers	Preservative	Analyses Required	Special Instructions for Lab
20180404	10:43:00	FL	04	0.1	1	SW	UKN				2 H HCl	TOC	
20180404	10:57:00	FL	03	0.1	1	SW	UKN				"	TOC	
20180404	11:12:00	FL	14	0.1	1	SW	UKN				"	TOC	
20180404	11:28:00	FL	15	0.1	1	SW	UKN				"	TOC	
20180404	11:53:00	FL	07	0.1	1	SW	UKN				"	TOC	
20180404	12:30:00	FL	01	0.1	1	SW	UKN				"	TOC	
20180404	13:08:00	FL	05	0.1	1	SW	UKN				"	TOC	
20180404	13:18:00	FL	02	0.1	1	SW	UKN				"	TOC	
20180405	14:15:00				1	DI	Hose DI				"	TOC	

* Matrix Codes: S
BI=Biotic Inver
SW=Surface W

Relinquished By:	Date:	Time:	Received By:	Date:	Time:	Remarks:

CHAIN OF CUSTODY FORM

(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

								Originator:			
								Ship Date:	April 5th 2018	Contact:	
								Destination:	Cincinnati	Phone #:	
								Recipient:	Sam Smith	E-Mail:	
								Phone #:	513-569-7681		
Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)		
20180403	12:42:00				1	DI	Field Blk.1	Chlorophyll	311.77		
20180403	12:44:00				1	DI	Field Blk.2	"	361.42		
20180404	08:38:00	FL	42	0.1	1	SW	UKN	"	137.07		
20180404	09:05:00	FL	44	0.1	1	SW	UKN	"	130.1		
20180404	09:36:00	FL	43	0.1	1	SW	UKN	"	151.73		
20180404	10:00:00	FL	41	0.1	1	SW	UKN	"	163.19		
20180404	10:43:00	FL	04	0.1	1	SW	UKN	"	163.16		
20180404	10:57:00	FL	03	0.1	1	SW	UKN	"	197.02		
20180404	11:12:00	FL	14	0.1	1	SW	UKN	"	274.25		
20180404	11:28:00	FL	15	0.1	1	SW	UKN	"	283.71		
20180404	11:53:00	FL	07	0.1	1	SW	UKN	"	266.18		
20180404	12:30:00	FL	01	0.1	1	SW	UKN	"	274.29		
20180404	13:08:00	FL	05	0.1	1	SW	UKN	"	274.29		
20180404	13:18:00	FL	02	0.1	1	SW	UKN	"	298.75		
		~~FL~~									

Relinquished By	Date	Time	Received By

Very Windy 4/4

Lots of Pollen! Smoke PM 4/5

GAS TRAP DATA SHEET

Deploy 4/4/18 Retrieve 4/5/18

Lake:						
Site	Lat	Long	Deployment time/date	Retrieval time/date	Volume recovered (mL)	Exetainer codes
53	36.069228	78.794079	8:30	8:17	1	
42	36.065337	78.795152	8:38	8:25	1	
50	36.068545	78.786124	8:51	8:34	1	
44	36.072960	78.779269	9:00	TRAP lost - Buoy detached		
52	36.079747	78.760392	9:15	9:02	+ TRAP lost - collector vessel sunk	
56	36.072877	78.766206	9:23	9:11	0	
47	36.068557	78.768534	9:29	9:18	0	
43	36.061512	78.773297	9:36	9:26	2	
54	36.057850	78.765112	9:45	9:38	0	
55	36.053192	78.761574	9:54	9:45	TRAP DAmaged	
41	36.055712	78.758627	10:00	9:53	2	
49	36.051257	78.756834	10:10	10:10	TRAP Damaged	
57	36.048507	78.752319	10:20	10:20	1	
45	36.044075	78.758207	10:27	10:30	6	
10	36.039793	78.756136	10:31	10:37	0	
4	36.012140	78.760904	10:43	10:49	0	
8	36.018825	78.748926	10:50	11:00	0	Lost Anchor Site ~200 m away
3	36.032318	78.736819	10:57	11:08	2	
14	36.059508	78.717217	11:12	11:24	0	
15	36.022667	78.714631	11:28	11:44	1	
11	36.019443	78.716991	11:37	12:09	30, 45	TRAP Moved to BANK 36.017605, 78.70454
7	36.005620	78.667749	11:53	12:23	0	
1	36.002138	78.634447	12:30	12:39	2	
21	36.008828	78.628139	12:37	12:48	0	
13	36.003620	78.646276	12:47	12:59	3	
12	35.965547	78.633862	13:00	13:13	0	
5	35.970070	78.621207	13:08	13:22	0	
2	35.976105	78.589029	13:18	13:35	2	
24	35.962043	78.588119	13:32	13:48	0	
16	35.956710	78.590486	13:40	14:00	0	

DISSOLVED GAS AND AIR DATA SHEET

Lake:	FAIS				Date:	4/5/18		
¥Site	Dissolved gas Exetainer codes		Water Vol/ Helium Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes			BP (mm Hg)
42	FI.42.DG.1	FI.42.DG.2	115/25	11.3				
44	FI.44.DG.1	FI.44.DG.2	↑	11.9				
41	FI.41.DG.1	FI.41.DG.2		13.5	FI.41.AA.1	FI.41.AA.2	FI.41.AA.3 FI.41.AA.4	760.9 FI.41.AA.5
43	FI.43.DG.1	FI.43.DG.2		12.9				
4	FI.04.DG.1	FI.04.DG.2		14.3				
3	FI.03.DG.1	FI.03.DG.2		14.9				
14	FI.14.DG.1	FI.14.DG.2		17.6				
15	FI.15.DG.1	FI.15.DG.2		14.3				
7	FI.07.DG.1	FI.07.DG.2		14.9				
1	FI.01.DG.1	FI.01.DG.2		15.2				
5	FI.05.DG.1	FI.05.DG.2	↓	16.6				BP = 760.9
2	FI.02.DG.1	FI.02.DG.2	115/25	16.8	FI.02.AA.1	FI.02.AA.2	FI.02.AA.3 FI.02.AA.4	FI.02.AA.5 Smoke (Heavy)

¥Site: all dissolved gas samples collected at 0.1m depth

Boat motor should be off for 2 minutes before sample collection

Goal is to inject 20mL of gas into exetainer

Signature Date

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES	
Lake:	FALLS
Sample Date:	6/5/18
GGA file	GGA-2018-06-05_f0000.txt
gps file name	

GPS Clock: 13:05:00 LGR Clock 14:03:29

DEPLOYMENT DATA

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L)	CH_4 and CO_2 profile checked	Water Temp (c)
53	08:17:10	✓	✓	50	Yes	14.0
42	08:24:51	✓	✓	50	✓	14.3
50	8:34:28	✓	✓	50	✓	14.3
44	8:48:50	✓	✓	50	✓	14.5
52	9:02:20	✓	✓	50	✓	15.5
56	9:11:22	✓	✓	50	✓	16.2
47	9:18:25	✓	✓	50	✓	15.4
43	9:28:14	✓	✓	50	✓	15.1
54	9:38:06	✓	✓	50	✓	15.3
55	9:45:56	✓	✓	50	✓	15.0
41	9:53:15	✓	✓	50	✓	15.0
49	10:10:15	✓	✓	45	✓	15.3
57	10:20:35	✓	✓	45	✓	15.5
45	10:30:15	✓	✓	45	✓	15.0
10	10:37:23	✓	✓	45	✓	15.3
4	10:48:57	✓	✓	45	✓	15.0
8	11:00:09	✓	✓	45	✓	15.8
3	11:08:33	✓	✓	45	✓	15.5
14	11:24:14	✓	✓	45	✓	15.8
15	11:44:00	✓	✓	50	✓	15.8
11	11:57:42	✓	✓	50	✓	14.8
7	12:23:45	✓	✓	45	✓	16.0
1	12:39:25	✓	✓	45	✓	17.0
21	12:48:45	✓	✓	45	✓	17.5
13	12:58:59	✓	✓	45	✓	16.9
12	13:13:50	✓	✓	45	✓	17.0
5	13:22:34	✓	✓	45	✓	17.6
2	13:35:27	✓	✓	45	✓	17.3
16	13:48:44	✓	✓	45	✓	16.5
24	14:00:33	✓	✓	45	✓	16.8

SONDE DATA SHEET

Lake: FALS Sample Date: 4/4/18

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
53	0.1	1.0	16.39	9.50	97.0	0.139	8.80	13.9	39.5	
42	0.1	1.3	16.61	9.50	97.3	0.160	8.79	12.6	35.0	
50	0.1	1.3	16.56	9.35	96.4	0.133	8.70	14.0	41.7	
44	0.1	1.3	16.66	9.37	96.4	0.117	8.71	12.0	44.8	
52	0.1	1.4	16.91	9.26	95.5	0.119	8.48	11.2	39.3	
56	0.1	2.5	16.84	9.26	95.2	0.119	8.39	8.8	43.8	
47	0.1	2.4	16.39	9.16	93.5	0.118	8.31	9.5	36.7	
43	0.1	1.9	16.56	9.57	98.0	0.115	8.19	10.0	38.1	
54	0.1	1.9	16.73	9.53	98.3	0.112	8.2	9.2	32.0	
55	0.1	2.4	16.48	9.42	96.3	0.114	8.2	10.6	28.5	
41	0.1	1.0	16.64	9.46	97.3	0.112	8.15	11.1	30.3	
49	0.1	2.9	16.23	9.41	95.2	0.114	8.12	9.3	30.6	
57	0.1	2.2	16.41	9.46	96.4	0.112	8.12	9.0	20.6	
45	0.1	2.6	16.65	9.62	99.2	0.113	8.08	8.1	18.8	
10	0.1	1.7	15.90	9.74	98.5	0.113	8.05	8.2	21.3	
4	0.1	1.2	15.45	9.15	91.7	0.130	8.02	9.3	25.9	
8	0.1	4.0	15.24	~~9.49~~ 9.55	95.2	0.128	8.00	6.4	15.8	
3	0.1	1.8	16.41	9.93	101.3	0.129	7.99	8.6	16.0	
14	0.1	1.7	16.15	9.60	97.3	0.140	7.97	5.4	19.0	
15	0.1	4.5	14.56	9.62	94.6	0.144	8.0	7.3	10.5	
11	0.1	3.1	14.27	9.20	90.2	0.146	7.96	4.9	11.1	
7	0.1	~~6.1~~ 8.0	15.58	10.47	106.1	0.143	8.04	~~27.1~~ 11.0	8.2	
7	1	~~6.1~~	14.68	10.43	102.4	0.143	8.00	~~33.0~~ 15.0	8.6	
7	2	~~6.1~~	14.49	10.23	99.9	0.142	7.98	~~32.0~~ 13.5	8.6	
7	3	~~6.1~~	13.88	~~10.31~~ 9.49	92.1	0.143	7.93	8.9	9.6	
7	4	~~6.1~~	12.88	~~9.00~~ 8.35	78.8	0.145	7.90	5.8	10.5	
7	5	~~6.1~~	12.47	7.76	~~72.8~~	0.142	7.87	5.9	9.7	
7	6	~~6.1~~	12.11	7.21	66.8	0.14,	7.83	6.3	~~9.7~~	
7	7	(7.6)	11.93	6.63	61.1	0.147	7.61	5.8	11.4	

Date 4/3/18 Pre-deployment

Cal cylinder serial # CC163816

Dilution system serial # EPA A20808

LGR file 559-2018-04-03_f0000.txt

Sample MFC gas correction factor 1.43

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.024	-0.01	13:59:26	14:02:10
7	70.0	28.3	283.2	25.93	265.1	14:02:10	14:06:33
6.5	70.0	30.5	304.7	27.78	286.0	14:06:33	14:09:20
6	70.0	33.0	329.8	30.03	310.6	14:09:20	14:11:37
5.5	70.0	35.9	359.4	32.70	339.4	14:11:37	14:14:33
5	70.0	39.5	394.9	35.80	373.7	14:14:33	14:18:03
4.5	70.0	43.8	438.1	39.50	414.9	14:18:03	14:28:21
4	70.0	49.2	491.9	44.21	466.2	14:28:21	14:30:36
3.5	70.0	56.1	560.8	49.8	531.9	14:30:36	14:33:01
3	70.0	65.2	652.1	57.58	619.6	14:33:01	14:36:10

Signature _____

Date 4/3/18
Cal cylinder serial # CC498857
Dilution system serial # EPA AZ0808
LGR file 994-2018-04-03.f0008.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.04	0.26	14:36:10	14:41:42
9	60.0	2.0	20.2	1.860	19.9	14:41:42	14:44:38
8.5	60.0	2.1	21.4	1.94	20.6	14:44:38	14:47:36
8	60.0	2.2	22.7	2.04	21.4	14:47:36	14:50:35
7.5	60.0	2.4	24.2	2.17	22.8	14:50:35	14:52:51
7	60.0	2.6	25.9	2.32	24.4	14:52:51	14:55:40
6.5	60.0	2.7	27.9	2.49	26.1	14:55:40	14:58:06
6	60.0	3.0	30.2	2.69	28.5	14:58:06	15:00:45
5.5	60.0	3.2	32.9	2.94	30.9	15:00:45	15:02:51
5	60.0	3.6	36.1	3.23	34.2	15:02:51	15:04:51

Signature _____

Date 4/8/18 Post-deployment

Cal cylinder serial # CC163816

Dilution system serial # EPA A20508

LGR file 339-2018-04-05_f0000.txt

Sample MFC gas correction factor 1.43

	Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time	
✓	3.25	0.0	0.0	0.0	0.030	0.340	10:39:00	10:40:45	
✓	7	70.0	28.3	283.2	25.58	260.08	10:40:45	10:42:42	
	6.5	70.0	30.5	304.7					
	6	70.0	33.0	329.8					
✓	5.5	70.0	35.9	359.4	32.45	337.5	10:43:40	10:44:09	
	5	70.0	39.5	394.9					
	4.5	70.0	43.8	438.1					
✓	4	70.0	49.2	491.9	44.00	464.8	10:44:09	10:48:25	
	3.5	70.0	56.1	560.8					
✓	3	70.0	65.2	652.1	~~59.46~~	~~599.0~~	10:48:25	10:50:40	Discard- Atm dump over a dilution system

Signature _____

Date 4/8/18

Cal cylinder serial # CC498857

Dilution system serial # EPA AZ0808

LGR file 394 - 2018-04-08-f0000.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.017	-0.01	10:55	10:59:43
9	60.0	2.0	20.2	1.86	20.2	10:59:43	11:01:57
8.5	60.0	2.1	21.4				
8	60.0	2.2	22.7	2.05	22.0	11:01:57	11:05:10
7.5	60.0	2.4	24.2				
7	60.0	2.6	25.9	2.33	24.9	11:05:10	11:08:07
6.5	60.0	2.7	27.9				
6	60.0	3.0	30.2	2.73	28.9	11:08:07	11:12:05
5.5	60.0	3.2	32.9				
5	60.0	3.6	36.1				

Signature _____

ATTACHMENT A
YSI 6600 SONDE CALIBRATION LOG SHEET

SONDE ID: STD 16 **DATE:** 4/4/18 **TECHNICIAN:** JTW

DATE / TIME OK? (YES) / NO EST

BATTERY VOLTS: 12.4 (Change if < 10.0 v) **CHANGED BATTERIES?** YES / (NO)

TURBIDITY WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

CHLOROPHYLL WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

DISSOLVED OXYGEN: "DO Sat %" **BAROMETRIC PRESSURE:** 746.2 mm Hg Post = 752.6 T = 21.0

SENSOR OUTPUT TEST: PASS / FAIL (For Rapid-Pulse Probes, change membrane if failed)

Record first 10 readings from Discrete Sampling:

_____ _____ _____ _____ _____

_____ _____ _____ _____ _____

DO MEMBRANE CHANGED? YES / (NO) **PROBE RE-SURFACED?** YES / (NO) **DATE/TIME:** _____

[~97] **INITIAL:** 97.5 **CALIBRATED:** 98.3 **POST-DEPLOYMENT:** 99.4

DO CHARGE [50 ± 25]: _____ **DO GAIN** [1.0 ± 0.35]: 1.05 206

DEPTH: "Pressure" [m] **PRESSURE OFFSET** [VENTED = 0 ± 6 / NON-VENTED = -14.7 ± 6]: 0.04963

[0] **INITIAL:** 0.046 **CALIBRATED:** 0.00 **POST-DEPLOYMENT:** -0.252

CONDUCTIVITY: "SpCond" [mS/cm] **CONDUCTIVITY CELL CONSTANT** [5.0 ± 0.45]: 4.96146

1.413 ~~[10]~~ **INITIAL:** 1.395 **CALIBRATED:** 1.413 **POST-DEPLOYMENT:** 1.413 T = 22.9

T = 23.0

TURBIDITY: "Turb+NTU" [NTU]

[0] **INITIAL:** 0.1 **CALIBRATED:** 0.0 **POST-DEPLOYMENT:** 0.1

126 ~~[1000]~~ **INITIAL:** 124.0 **CALIBRATED:** 126.0 **POST-DEPLOYMENT:** 124.3

CHLOROPHYLL: "Chlor ug/L" [ug/L]

[0] **INITIAL:** 0.0 **CALIBRATED:** _____ **POST-DEPLOYMENT:** 0.1 ± 0.3

~~[113]~~ **INITIAL:** _____ **CALIBRATED:** _____ **POST-DEPLOYMENT:** 12.4 Site 2 T = 7.37

Post-Deployment: ~~11.6~~ 11.6 Site 42 T = 6.50

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pH: "pH"

[7] **INITIAL:** 6.98 **CALIBRATED:** 7.0 **POST-DEPLOYMENT:** 6.99 **mV** [0 ± 40]: -37.7 26.98

[10] **INITIAL:** 9.95 **CALIBRATED:** 10.0 **POST-DEPLOYMENT:** 9.98 **mV** [-177 ± 40]: -211.0 @ 10.0

ORP: "ORP mV" **ORP mV OFFSET [0 ± 100]:** _____